

Karan Shihire

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SUMMARY

- AI Engineer specializing in retrieval systems and applied NLP — building end-to-end ML applications from embedding pipelines to deployed APIs. Hands-on with semantic search, FAISS vector retrieval, and transformer embeddings.

PROJECTS

- **CareerLens — Semantic Job Recommendation Engine** [GitHub](#)
 - Built a Retrieval-Augmented Generation (RAG) pipeline using Sentence Transformers, FAISS, and the OpenAI API to retrieve semantically relevant job descriptions and generate personalized career recommendations.
 - Designed a hybrid ranking pipeline combining embedding similarity, FAISS retrieval, and skill-overlap reranking to surface why each job was recommended, not just a similarity score.
 - Implemented a modular FastAPI backend (5 services: parsing, embedding, retrieval, recommendation scoring, API) handling 88 requests/sec at 95ms average latency.
 - Deployed the FastAPI backend on Railway and the React frontend on Netlify with automated GitHub CI/CD, environment-based configuration, and CORS support for production.
- **FaceTrack — Real-Time Face Recognition / Attendance Platform** [GitHub](#)
 - Built a real-time face recognition system (InsightFace + FAISS) to identify multiple enrolled individuals from live webcam streams.
 - Designed a multi-angle face enrollment pipeline to improve recognition robustness across different poses and lighting conditions.
 - Challenge: matching faces reliably across lighting changes — solved using multi-angle enrollment and FAISS cosine similarity threshold tuning.
 - Architected the system into 5 modular components: detection, embedding, vector search, recognition, and backend services.
- **NetWatch — Anomaly-Based Intrusion Detection System (IDS)** [GitHub](#)
 - Developed a real-time anomaly detection system that monitors live network traffic and identifies suspicious behavior using an Isolation Forest model.
 - Engineered a Scapy-based packet capture pipeline to extract network features from TCP, UDP, and ICMP traffic for real-time ML inference.
 - Built a Flask dashboard for visualizing packet statistics, anomaly alerts, and live monitoring results.
 - Designed a concurrent pipeline separating packet capture, feature extraction, anomaly prediction, and dashboard updates.

TECHNICAL SKILLS

Languages: Python, SQL, Java

AI / ML: PyTorch, Scikit-learn, Transformer Embeddings, Retrieval-Augmented Generation (RAG), Vector Similarity Search, Semantic Search, FAISS, NLP, NumPy, Pandas

Backend / Database: FastAPI, Flask, REST API, PostgreSQL, Railway

Deployment / Tools: Docker, GitHub, Render, Netlify, HuggingFace

EDUCATION

Modern College of Arts, Science and Commerce

Bachelor of Computer Science (Minor: Embedded Systems & IoT) | CGPA: 8.0

Pune, Maharashtra

Jul 2023 – May 2026

CERTIFICATIONS

Microsoft Azure Machine Learning Fundamentals

Machine Learning Concepts, Azure ML Services, Responsible AI

Microsoft

Oracle Cloud Infrastructure Foundations Associate

Cloud Architecture, OCI Core Services, Identity & Security

Oracle